FIITJEE

# JEE(Advanced)-2018

ALL INDIA TEST SERIES

## ANSWERS, HINTS & SOLUTIONS

## PART TEST – III

### PAPER-2

| Q. No. | PHYSICS | Q. No. | CHEMISTRY | Q. No. | MATHEMATICS |
|---|---|---|---|---|---|
| 1. | **A, C, D** | 21. | **A, C, D** | 41. | **B, C** |
| 2. | **B, D** | 22. | **B, C, D** | 42. | **A, B, C** |
| 3. | **B, C** | 23. | **B, C** | 43. | **A, B, C** |
| 4. | **B, C, D** | 24. | **A, B, C, D** | 44. | **A, B, C** |
| 5. | **B, D** | 25. | **A, B** | 45. | **A, B** |
| 6. | **A, C** | 26. | **C** | 46. | **A, C, D** |
| 7. | **B, C** | 27. | **A, B, C, D** | 47. | **A, C, D** |
| 8. | **A, C** | 28. | **C** | 48. | **A, B, C** |
| 9. | **C** | 29. | **C** | 49. | **A** |
| 10. | **B** | 30. | **B** | 50. | **D** |
| 11. | **D** | 31. | **D** | 51. | **D** |
| 12. | **B** | 32. | **B** | 52. | **C** |
| 13. | **3** | 33. | **6** | 53. | **5** |
| 14. | **5** | 34. | **1** | 54. | **5** |
| 15. | **5** | 35. | **2** | 55. | **8** |
| 16. | **2** | 36. | **1** | 56. | **0** |
| 17. | **4** | 37. | **3** | 57. | **1** |
| 18. | **6** | 38. | **4** | 58. | **8** |
| 19. | **5** | 39. | **0** | 59. | **7** |
| 20. | **8** | 40. | **5** | 60. | **6** |

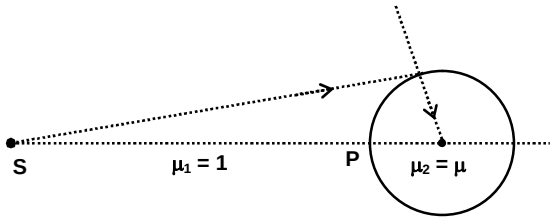
# Physics

## PART – I

### SECTION – A

1. (A) $\frac{\mu}{+R} = \frac{\mu - 1}{+R}$ (not possible)

The higher the refractive index of the material of sphere ($\mu$), the closer one could come to fulfilling the objective




(B) $\frac{f_a}{f_w} = \left(\frac{\mu - \mu_w}{\mu - \mu_a}\right)\left(\frac{\mu_a}{\mu_w}\right)$

$= \left(\frac{\frac{3}{2} - \frac{4}{3}}{\frac{3}{2} - 1}\right)\frac{1}{\left(\frac{4}{3}\right)} = \frac{1}{6} \times 2 \times \frac{3}{4} = \frac{1}{4}$

(C) basic (NCERT-point to ponder)

(D) basic (NCERT-point to ponder)

2. As we know that,

$\frac{hC}{\lambda} = \phi + K \Rightarrow \frac{12400}{\lambda} = \phi + v_s$, where $\phi$ is in electron Volt and $\lambda$ is in Å

When light of wavelength $\lambda$ = 6000 Å is used

$\frac{12400}{6000} = \phi + 0.3 \Rightarrow \phi$ = 1.77 ≈ 1.8 eV (approximately)

When light with wavelength $\lambda$ = 4000 Å is used

$\frac{12400}{4000} = \phi + eV$

$\Rightarrow$ eV = (3.1 – 1.8) e volt = 1.3 eV

$\Rightarrow$ v = 1.3 Volt

3. Condition for maximum intensity due to thin film

$2\mu d - \frac{\lambda}{2} = n\lambda$

$\Rightarrow 2\mu d = (2p+1)\frac{\lambda_1}{2} \Rightarrow (2p + 1)\lambda_1 = 4\ \mu d$

Here $\lambda_1$ = 400 nm and n = p $\Rightarrow$ An integer

Let $\lambda'$ = 1150 nm. Since only two wavelength give maximum intensity, so

$\frac{4\mu d}{2(p-1)+1} < \lambda' < \frac{4\mu d}{2(p-2)+1}$

$\frac{\lambda_1(2p+1)}{2(p-1)+1} < \lambda' < \frac{\lambda_1(2p+1)}{2(p-2)+1}$

$\Rightarrow \frac{\lambda_1(2p+1)}{2(p-1)+1} < \lambda'$ or $\lambda' < \frac{\lambda_1(2p+1)}{2(p-2)+1}$

$\Rightarrow p > \frac{1}{2}\left(\frac{\lambda' + \lambda_1}{\lambda' - \lambda_1}\right) = \frac{1}{2}\left(\frac{1150 + 400}{1150 - 400}\right) = 1.....$, or

$$p < \frac{1}{2}\left(\frac{3\lambda' + \lambda_1}{\lambda' - \lambda_1}\right) = \frac{1}{2}\left(\frac{3\times 1150 + 400}{1150 - 400}\right) = 2.....$$

The only integer which satisfies both inequalities is 2, so

$$d = \frac{(2\times 2 + 1)400\times 10^{-9}}{4\times 1} = 500 \text{ nm}$$

$$\lambda_2 = \frac{4\mu d}{2(p-1)+1} = \frac{4\times 1\times 500\times 10^{-9}}{2+1} = 666.7 \text{ nm}$$

$$\Delta T = \frac{d}{\alpha h} = \frac{500\times 10^{-9}}{8\times 10^{-6}\times 2\times 10^{-2}} \approx 3.1\,^\circ C$$

4. $k$ = stiffness of string $= \frac{F}{x_{max}} = \frac{mg}{x_{max}} = \frac{40\times 10^{-3}\times 9.8}{20\times 10^{-2}} = 1.96 \text{ N/m}$

When we replace the mass 40 gm by the mass $m_2 = \frac{2205}{32}$ gm, we have

$$T = 2\pi\sqrt{\frac{m}{k}} = \frac{3\pi}{8} \text{ sec}$$

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}} = \frac{16}{3} \text{ rad/sec}$$

$$x = x_0 \cos \omega t = 0.2 \cos \frac{16}{3}t$$

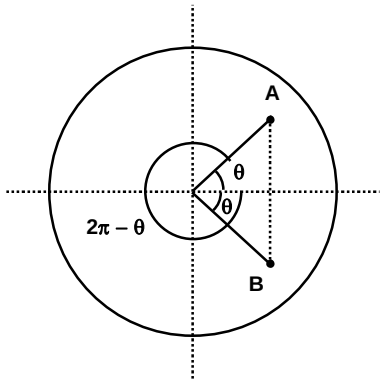
$$v = -x_0\omega \sin \omega t = -\frac{16}{15}\sin\frac{16}{3}t$$

$$a = -x_0\,\omega^2 \cos \omega t = -\frac{256}{45}\cos\frac{16}{3}t$$

If $x = 10$ cm $\Rightarrow \cos\frac{16}{3}t = \frac{1}{2} = \cos\left(\frac{\pi}{3}\right)$

$$\Rightarrow \frac{16}{3}t = (2n\pi \pm \frac{\pi}{3})$$

$$\Rightarrow t = \left(\frac{3n\pi}{8} \pm \frac{\pi}{8}\right)\text{sec}$$




5. L.C. $= \frac{1}{20}\times 0.1 = 0.005$ cm

Let 1 V.S.D. = x

$$\Rightarrow \frac{1}{20} = 1 - x \Rightarrow x = 0.95 \text{ mm}$$

$$(n-4)\times 1 = n\times 0.95$$

$$\Rightarrow n = 80$$

Length of pencil $= 8.4 + 0.005\times\left(\frac{80}{5}\right) = 8.48$ cm

6. $\phi = \frac{2\pi}{\lambda}\delta$

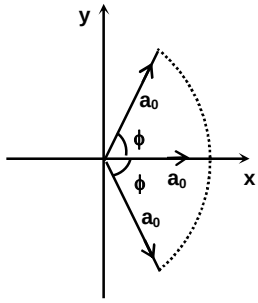
$y_2 = a_0 \sin(\omega t - kx)$

$\Rightarrow y_1 = a_0 \sin(\omega t - kx - \phi)$ and $y_3 = a_0 \sin(\omega t - kx + \phi))$

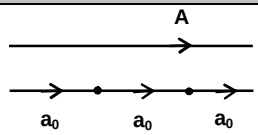
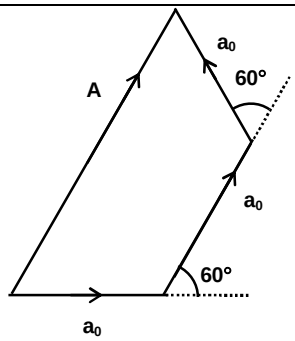
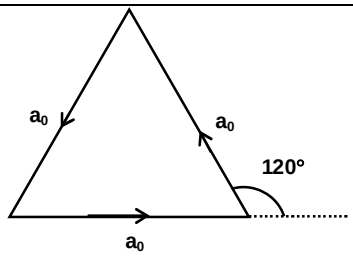
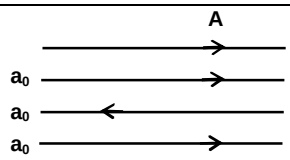
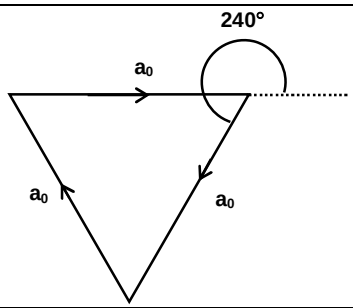
$$A = |a_0 + 2a_0\cos\phi| = a_0|(1+2\cos\phi)|$$

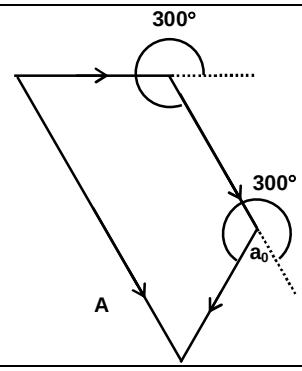
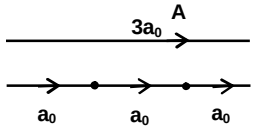
$$\Rightarrow A = a_0|(1+2\cos\left(\frac{2\pi xd}{\lambda D}\right)|$$

$$= a_0|(1+2\cos\left(\frac{2\pi\delta}{\lambda}\right)|$$




**Second Method :** Suppose $\phi$ and $\delta$ represent the phase difference and path difference between two connective waves so

| $\phi$ | $\delta$ | Phasor representation | Resultant amplitude |
|---|---|---|---|
| 0 | 0 |   | $A = 3a_0$ |
| 60° | $\frac{\lambda}{6}$ |   | $A = 2a_0$ |
| 120° | $\frac{\lambda}{3} = \frac{2\lambda}{6}$ |   | A = 0 |
| 180° | $\frac{\lambda}{2} = \frac{3\lambda}{6}$ |   | $A = a_0$ |
| 240° | $\frac{2\lambda}{3} = \frac{4\lambda}{6}$ |   | A = 0 |

| | | | |
|---|---|---|---|
| 300° | $\frac{5\lambda}{6}$ |   | A = $2a_0$ |
| 360° | $\lambda = \frac{6\lambda}{6}$ |   | A = $3a_0$ |

7. For transverse wave, $v_T = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{F}{\rho \times S}}$

$$V_{Al} = \sqrt{\frac{900}{2.5\times10^3\times10^{-6}}} = \frac{30}{50\times10^{-3}} = 600\ m/s$$

$$V_{Cu} = \sqrt{\frac{900}{8.1\times10^3\times10^{-6}}} = \frac{30}{90\times10^3} = \frac{1000}{3} m/s$$

$$\frac{a_r}{a_i} = \frac{V_{Al}-V_{Cu}}{V_{Al}+V_{Cu}} = \frac{\frac{800}{3}}{\frac{2800}{3}} = \frac{2}{7} \approx 0.3$$

For longitudinal wave $V_\ell = \sqrt{\frac{Y}{\rho}}$

$$V_{Al} = \sqrt{\frac{9\times10^{10}}{2.5\times10^3}} = \frac{3\times10^5}{50} = 6000\ m/s$$

$$V_{Cu} = \sqrt{\frac{16\times10^{10}}{8.1\times10^3}} = \frac{4\times10^5}{9\times10} = \frac{40000}{9}\ m/s$$

$$\frac{a_r}{a_i} = \frac{V_{Al}-V_{Cu}}{V_{Al}+V_{Cu}} = \frac{\frac{1400}{9}}{\frac{94000}{9}} = \frac{7}{47} \approx 0.15$$

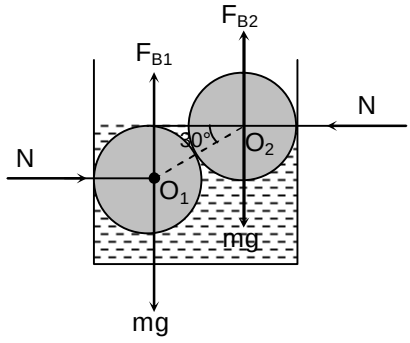
8. $2mg = F_{B1} + F_{B2}$

$\Rightarrow \rho_{wood} = \frac{3}{4}\rho_{water} = 750 kg/m^3$

Taking torque about $O_1$

$N\times R = (mg - F_{B2})\times 2R\cos30°$

$\Rightarrow = 45\times10\times\frac{1}{3}\times1.732 = 259.8 \approx 260 N$



9. During one time period ($T_0$)

$$W = \frac{T\omega^2 y_0^2}{2v}\left[\int_0^{T_0}\left(1+\cos\left\{2\omega\left(t-\frac{x}{v}\right)\right\}\right)dt\right]$$

Average Power = $\frac{W}{T_0} = \frac{\omega^2 y_0^2 T}{2v}$

10. Since the work done will be same at any place along the string in one cycle so choose x = 0, for convenience. Then work done in a cycle = $\frac{\omega^2 y_0^2 T}{v}\int_0^{2\pi/\omega}\cos^2\omega t dt = \frac{\pi\omega y_0^2 T}{v}$

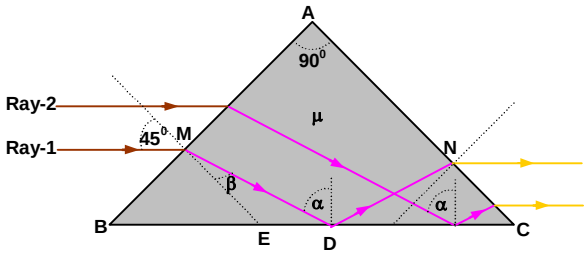
Average power = work done per unit time = $\frac{\text{work done in a cycle}}{\text{Time period}} = \frac{\frac{\pi\omega y_0^2 T}{v}}{\frac{2\pi}{\omega}} = \frac{\omega^2 y_0^2 T}{2v}$

Average power = $2\pi^2 f^2 y_0^2 T\sqrt{\frac{\mu}{T}} = 2\pi^2 f^2 y_0^2\sqrt{\mu T}$

$= 2\times10\times200\times200\times0.2\times0.2\times10^{-4}\sqrt{5\times10^{-4}\times5}$ = 160 × $10^{-3}$ Watt = 0.16 Watt.

11.-12. In figure-I, total internal reflection takes place at face AB.

$\angle EMD + \angle MDE = \angle BEM$




$\Rightarrow \beta + (90^\circ - \alpha) = 45^\circ$

$\Rightarrow \sin(\beta + 45^\circ) > \left(\frac{1}{\mu}\right)$

$\Rightarrow \sin\beta + \cos\beta > \frac{\sqrt{2}}{\mu}$ .....( 1)

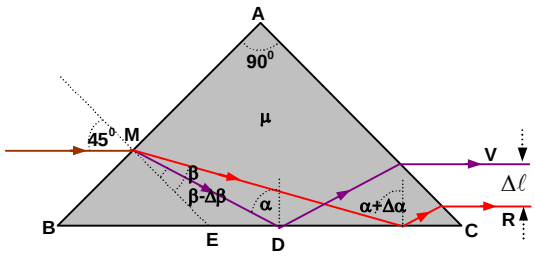
According to law of refraction at point M, we can write

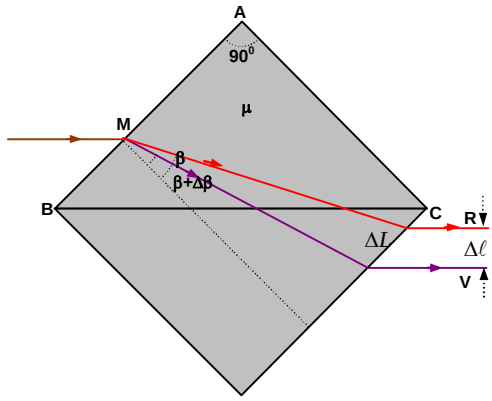
$1\times\sin(45^\circ) = \mu\sin\beta \Rightarrow \sin\beta = \frac{1}{\mu\sqrt{2}} \Rightarrow \cos\beta = \sqrt{1-\frac{1}{2\mu^2}}$

Putting these value in equation ( 1), we have

$\frac{1}{\mu\sqrt{2}} + \sqrt{1-\frac{1}{2\mu^2}} > \frac{\sqrt{2}}{\mu} \Rightarrow \sqrt{2\mu^2-1} > 1 \Rightarrow \mu > 1$

**In figure-II:** Replace the **Dove Prism,** by glass cube of side as AB.



$$\Delta\ell = \frac{\Delta L}{\sqrt{2}} \text{ ……..( 2)}$$

From geometry, we can write

$$\sin\beta = \frac{1}{\mu_R\sqrt{2}} \text{ and } \sin\left(\beta-\Delta\beta\right) = \frac{1}{\left(\mu_R+\Delta\mu_R\right)\sqrt{2}}$$

$$\Delta L = a\tan\beta - a\tan\left(\beta-\Delta\beta\right) = \frac{a}{\sqrt{2\mu_R^2-1}} - \frac{a}{\sqrt{2\left(\mu_R+\Delta\mu\right)^2-1}}$$

$$\Rightarrow \Delta L = \frac{a}{\sqrt{2\mu_R^2-1}}\left(1-\sqrt{\frac{2\mu_R^2-1}{2\left(\mu_R+\Delta\mu\right)^2-1}}\right) = \frac{a}{\sqrt{2\mu_R^2-1}}\left(1-\sqrt{\frac{2\mu_R^2-1}{2\left(\mu_R+\Delta\mu\right)^2-1}}\right)$$

$$\Rightarrow \Delta L = \frac{a}{\sqrt{2\mu_R^2-1}}\left(1-\sqrt{\frac{2\mu_R^2-1}{2\mu_R^2+4\mu_R\Delta\mu+2\left(\Delta\mu\right)^2-1}}\right)$$

Neglecting the value $\left(\Delta\mu\right)^2$, we have

$$\Rightarrow \Delta L = \frac{a}{\sqrt{2\mu_R^2-1}}\left(1-\sqrt{\frac{1}{1+\frac{4\mu_R\Delta\mu}{2\mu_R^2-1}}}\right) = \frac{a}{\sqrt{2\mu_R^2-1}}\left(1-\left\{1+\frac{4\mu_R\Delta\mu}{2\mu_R^2-1}\right\}^{-\frac{1}{2}}\right)$$

Expand binomially, we have

$$\Rightarrow \Delta L = \frac{2a\mu_R\Delta\mu}{\left(2\mu_R^2-1\right)^{\frac{3}{2}}}$$

Putting this value in equation ( 2), we have

$$\Delta\ell = \frac{\sqrt{2}a\mu_R\Delta\mu}{\left(2\mu_R^2-1\right)^{\frac{3}{2}}}$$

**Second method:**

$$\Delta L = \left(\frac{d}{d\beta}\left(a\tan\beta\right)\right)\times\Delta\beta$$

$$\Delta L = \left(a\sec^2\beta\right)\times\Delta\beta$$

Also $\sin\beta = \frac{1}{\sqrt{2}\mu_R}$

$$\Rightarrow \quad \cos\beta\Delta\beta = \frac{1}{\sqrt{2}\mu_R^2}\Delta\mu$$

$$\Rightarrow \quad \Delta L = \frac{a\sec^2\beta\times\Delta\mu}{\sqrt{2}\mu_R^2\cos\beta}$$

$$\Rightarrow \quad \Delta L = \frac{2a\mu_R\Delta\mu}{\left(2\mu_R^2-1\right)^{\frac{3}{2}}}$$

**Third method:**

$$\Delta\ell = \frac{\Delta L}{\sqrt{2}},\ \sin\alpha = \frac{1}{\mu_r\sqrt{2}},\ \sin\beta = \frac{1}{\mu_v\sqrt{2}}$$

$$\Delta L = a\tan\alpha - a\tan\beta$$

$$= a\left[\frac{1}{\sqrt{2\mu_r^2-1}} - \frac{1}{\sqrt{2\mu_v^2-1}}\right]$$

$$= a\left[\frac{1}{2} - \frac{1}{2\sqrt{1+\mu_r\Delta\mu}}\right]$$

$$= \frac{a}{2}\left[1-\left(1+\mu_r\Delta\mu\right)^{-1/2}\right]$$

$$= \frac{a}{2}\left[1-1+\frac{1}{2}\mu_r\Delta\mu\right]$$

$$= \frac{a}{4}\mu_r\Delta\mu$$

$$\Delta\ell = \frac{\Delta L}{\sqrt{2}} = \frac{a}{4\sqrt{2}}\mu_r\Delta\mu = \frac{4\,cm}{4\sqrt{2}}\times\frac{\sqrt{5}}{\sqrt{2}}\times\frac{2}{100}$$

$$= \frac{\sqrt{5}}{100}cm = \sqrt{5}\times10^{-4}\,m$$

$$= 2.25\times10^{-4}\,m$$

Since: $\Delta\ell \geq d$

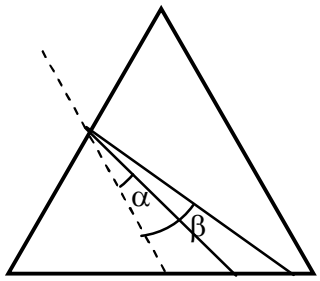
$$\Rightarrow \quad d \leq 2.25\times10^{-4}\,m$$

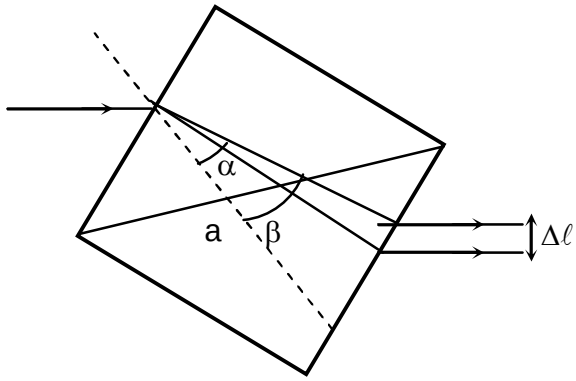
$$\sqrt{2\mu_v^2-1} = \sqrt{2\left(\mu_r+\Delta\mu\right)^2-1}$$

$$= \sqrt{2\mu_r^2+2\left(\Delta\mu\right)^2+4\mu_r\Delta\mu-1}$$

$$= \sqrt{4+4\mu_r\Delta\mu}\ \left[\left(\Delta\mu\right)^2 \approx 0\right]$$

$$= 2\sqrt{1+\mu_r\Delta\mu}$$




**SECTION – C**

13. $\frac{EB}{\mu} = \frac{B^2C}{\mu}$ = energy density × speed

= Energy per unit volume × speed

= $\frac{\text{Energy}}{\text{volume}}\times$ speed

$$\frac{ML^2T^{-2}}{L^3}\times LT^{-1} = ML^0T^{-3}$$

Here a = 1, b = 0 c = –3 $\Rightarrow \frac{b-c}{a} = \frac{0-(-3)}{1} = 3$

14. $\mu$ = mass per unit length = $\frac{m}{\ell} = \frac{0.6\times10^{-3}}{2} = 3 \times 10^{-4}$ kg/m

T = mg = 0.5 $\times$ 9.8 = 4.9 N $\Rightarrow$ Tension in the string

v = speed of wave in string = $\sqrt{\frac{T}{\mu}}$

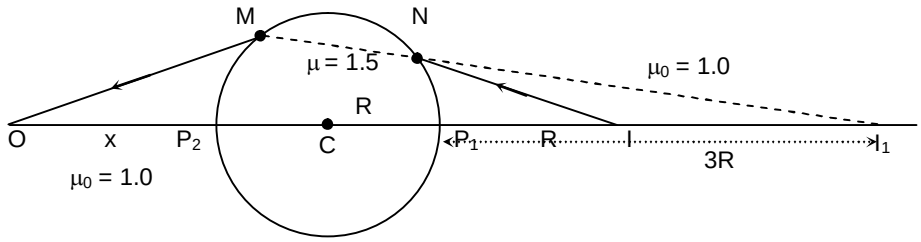
= $\sqrt{\frac{4.9}{3\times10^{-4}}} = 128$ m/s

$\lambda \Rightarrow$ wavelength of the wave = $\frac{v}{f} = \frac{128}{400} = 32\times10^{-2}\,m = 2^5 \times 10^{-2}$ m

$\Rightarrow n = 5$

15. We can reverse the question. Let I behaves as object and O behaves as image, because retracing the path will not alter the distances.
Consider first refraction at point N

$\frac{1.5}{v} - \frac{1.0}{-R} = \frac{1.5-1.0}{+R} \Rightarrow \frac{1.5}{v} = \frac{1}{2R} - \frac{1}{R} \Rightarrow v = -3R$




Consider second refraction at point M, for this I, will behaves as normal object so

$\frac{1.0}{v} - \frac{1.5}{-5R} = \frac{1.0-1.5}{-R}$

$\Rightarrow \frac{1}{v} + \frac{3}{10R} = \frac{1}{2R} = \frac{5}{10R}$

$\Rightarrow \frac{1}{v} = \frac{5}{10R} - \frac{3}{10R} = \frac{1}{5R}$

$\Rightarrow v = 5R$

$\Rightarrow x = 5R \Rightarrow k = 5$

16. We have to calculate the partial pressure, generated due the liquid of variable density

$\rho = \frac{\rho_0}{H}y + \rho_0$

W = weight of liquid column above the point A

$\Rightarrow W = \int_y^H S\rho g dy = gS\rho_0 \int_y^H \left(1+\frac{y}{H}\right)dy$

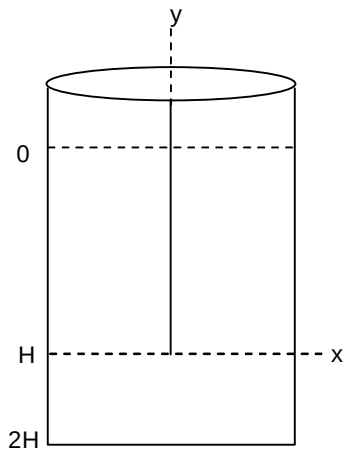
$= gs\rho_0\left[(H-y) + \frac{H^2-y^2}{2H}\right] = \frac{S\rho_0(H-y)(3H-y)g}{2H}$

$P_1$ = Partial pressure at interface (y = 0) due to liquid with variable density = $\left.\frac{W}{S}\right|_{y=0}$

$= \frac{3\rho_0 gH}{2}$

P $\Rightarrow$ Pressure at bottom = $P_0 + P_1 + \rho_0 gH = \rho_0 + \frac{5\rho_0 gH}{2}$

$= 10^5 + \frac{5\times10^3\times10\times4}{2} = 2\times10^5 = 2$ atm




17. Use Bernoulli's equation at point-3 and point-4, we can write

$\frac{P_0 + \Delta P}{\rho_{air}} = \frac{P_{BC}}{\rho_{air}} + \frac{1}{2}v^2 \Rightarrow P_{BC} = P_0 + \Delta P - 0.65\, v^2$

Using Bernoulli's equation at point -1 and point -2, we can write

$\frac{P_1}{\rho_{liquid}} + \frac{1}{2}v_{liquid}^2 = \frac{P_2}{\rho_{liquid}}$

$\Rightarrow \frac{P_{BC} + \rho_{\ell iquid} gh}{\rho_{\ell iquid}} + \frac{1}{2}v_\ell^2 = \frac{P_0}{\rho_\ell}$

$\Rightarrow \frac{P_0 + \Delta P + \rho_\ell gh - 0.65v^2}{\rho_\ell} + \frac{1}{2}v_\ell^2 = \frac{P_0}{\ell}$

$\Rightarrow \frac{1}{2}\rho_\ell v_\ell^2 = 0.65v^2 - \Delta P - \rho_\ell gh \geq 0$

$\Rightarrow v^2 \geq \frac{\Delta P + \rho_\ell gh}{0.65}$

$v_{min} = \sqrt{\frac{\Delta P + \rho_\ell gh}{0.65}} = \frac{100}{5} = 20$ m/s

18. According to Bohr's model

$\Delta E = E_0\left(1 - \frac{1}{n^2}\right) \Rightarrow \Delta E_{min} = \frac{3E_0}{4}$

During inelastic collision, a part of kinetic energy of colliding particles is converted into internal energy. The internal energy of the system of two hydrogen atoms, considered in the problem cannot be changed less than $\Delta E_{min}$. It means if the change in kinetic energy of system in ground frame is less than $\Delta E_{min}$ (or if the kinetic energy of colliding atoms with respect to their centre of mass is less than $\Delta E_{min}$), then collision must be an elastic one. Hence considering the critical case

$\frac{1}{2}m_H\left(\frac{v_0}{2}\right)^2 \times 2 = \frac{3E_0}{4} \Rightarrow v_0 = \sqrt{\frac{3E_0}{m_H}}$

$\sqrt{\frac{3\times2.18\times10^{-18}}{1.67\times10^{-27}}} = \sqrt{39.1617}\times10^4 = 6.257\times10^4\ m/s \approx 6.26\times10^4\ m/s$

$\frac{a\times b\times c}{k\times n} = \frac{6\times2\times6}{3\times4} = 6$

19. $\cos\theta = \frac{r}{R} = \frac{1}{2}$

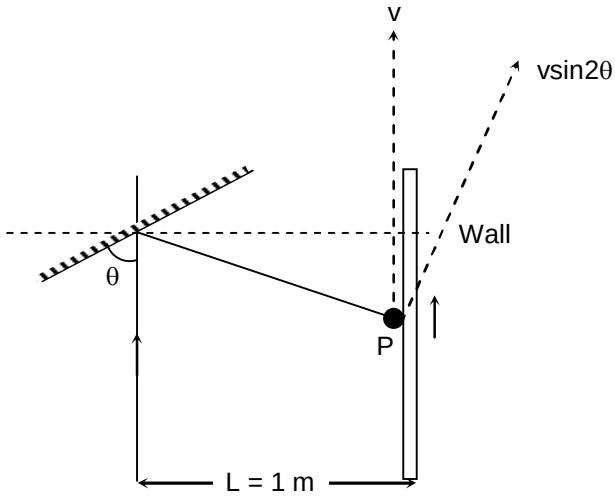
$4\pi rT\cos\theta = \pi r^2 \Delta P$

20. $\tan(2\theta - 90) = \frac{Y}{L}$

$y = -L\cot 2\theta$

$\frac{dy}{dt} = -L \times (-\text{cosec}^2 2\theta) \times 2\frac{d\theta}{dt}$

$= 2L(\text{cosec}^2 2\theta)\frac{d\theta}{dt}$



# Chemistry

## PART – II

### SECTION – A

21. (A) $2CrI_3 + 27Cl_2 + 64KOH \longrightarrow 2K_2CrO_4 + 54KCl + 6KIO_4 + 32H_2O$

(B) n-Factor of $(NH_4)_2Cr_2O_7$ is 6.

(C) $\left[\overset{+3}{Fe}\left(\overset{+2\ -3}{CN}\right)_6\right] \longrightarrow Fe^{3+} + \overset{+4}{C}O_2 + \overset{+5}{N}O_3^-$

n-Factor = 60.

(D) $\underset{n-Factor=\frac{2}{3}}{\overset{+6}{Mn}O_4^{2-}} + H^+ \longrightarrow \overset{+7}{Mn}O_4^- + \overset{+4}{Mn}O_2$

Hence, eq. wt. of $MnO_4^{2-} = \dfrac{119 \times 3}{2} = 178.5$

22. B. $CuCl_2 + H_2S \longrightarrow CuS + 2HCl$

C. $\left[Cd(CN)_4\right]^{2-} + H_2S \longrightarrow CdS + 2H^+ + 4CN^-$

D. $\left[Zn(OH)_4\right]^{2-} + H_2S \longrightarrow ZnS + 2OH^- + 2H_2O$

23.

| | $2A \rightleftharpoons A_2$ | |
|---|---|---|
| moles at $t = 0$ | m | 0 |
| moles at $t_{eq}$ | $m - m\alpha$ | $\dfrac{m\alpha}{2}$ |

Total moles = $m - \dfrac{m\alpha}{2} = m\left(1 - \dfrac{\alpha}{2}\right)$

$\Delta T_b = iK_bm = K_bm\left(1 - \dfrac{\alpha}{2}\right)$

$1 - \dfrac{\alpha}{2} = \dfrac{\Delta T_b}{K_bm}$

$\alpha = \dfrac{2(K_bm - \Delta T_b)}{K_bm}$

Equilibrium constant K for the dimer formation is

$K = \dfrac{[A_2]}{[A]^2} = \dfrac{m\alpha}{2[m(1-\alpha)]^2}$

$K = \dfrac{\alpha}{2m(1-\alpha)^2}$

$K = \dfrac{2(K_bm - \Delta T_b)}{K_bm} \Big/ 2m\left[1 - \dfrac{2(K_bm - \Delta T_b)}{K_bm}\right]^2$

$K = \dfrac{K_b(K_bm - \Delta T_b)}{(K_bm - 2K_bm + 2\Delta T_b)^2}$

$K = \dfrac{K_b(K_bm - \Delta T_b)}{(2\Delta T_b - K_bm)^2}$

24. Cloud seeding is a form of weather modification, a way changing the amount or type of precipitation that falls from clouds by dispersing chemicals into the air.

25. $Fe_2O_3$, $\phi = \frac{\pi}{3\sqrt{2}}\left\{1+\frac{2}{3}(0.414)^3\right\}$

Diamond, $\phi = 8\times\frac{4\pi r^3}{3\times\left(2\sqrt{2}r\right)^3}$

26. Diaspore is an aluminium oxide hydroxide mineral crystallizing in the orthorhombic system.

28. A represent liquid composition and B represent vapour composition.

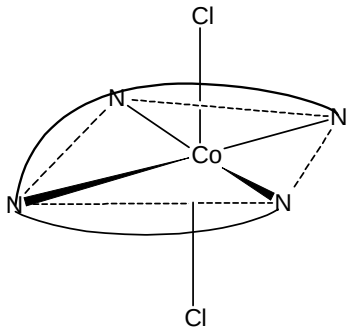
29. Above packing represent two possible arrangements, hexagonal close packing or cubic close packing, hence minimum distance, a or $\frac{a}{\sqrt{2}}$.

30. Above information represent cubic close packing

$$\rho = \frac{197\times 4}{(4.07)^3\times 10^{-24}\times 6.02\times 10^{23}}$$

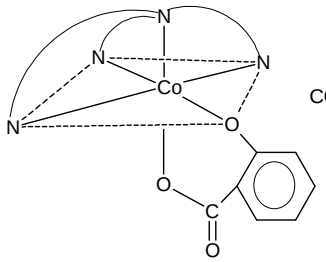
$\rho = 19.41\ g/cc$

31. (A)

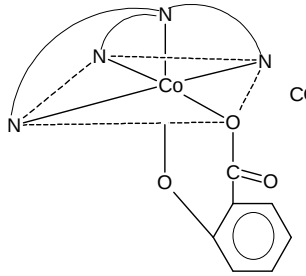



*trans* form has 3 coplanar ring

(C)



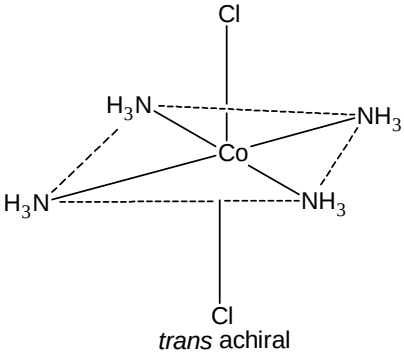

$COO^-$ *trans* to tertiary N



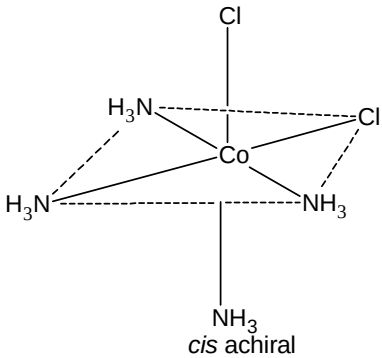

$COO^-$ cis to tertiary N

$[Pt(NH_3)_2(py)_2Cl_2]^{2+}$ has 4 non chiral isomer.

32.




*trans* achiral




*cis* achiral

It has axis of symmetry and plane of symmetry

**SECTION – C**

33. The energy involved during oxidation of glucose is $= -\frac{2880 \times 150}{180} = -2400$ kJ

Energy available for muscular work $= \frac{2400}{4} = 600$kJ

Energy required for 1 km walk = 100 kJ

Therefore the distance that a person will be able to walk taking 150 g glucose = 6 km.

34. $14H^+ + Cr_2O_7^{--} + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$

$$E_{Cr_2O_7^{--}/Cr^{3+}} = E^o_{Cr_2O_7^{--}/Cr^{3+}} - \frac{0.059}{6}\log_{10}\frac{\left[Cr^{3+}\right]^2}{\left[Cr_2O_7^{--}\right]\left[H^+\right]^{14}}$$

$$E_{Cr_2O_7^{--}/Cr^{3+}} = 1.33 + \frac{0.059}{6}\log_{10}\frac{\left(\frac{4.5}{1000}\right)\left(10^{-2}\right)^{14}}{\left(\frac{15}{1000}\right)^2}$$

$$= 1.33 + \frac{0.059}{6}\log_{10}\frac{4.5\times10^{-31}}{2.25\times10^{-4}}$$

$$= 1.33 + \frac{0.059}{6}\log_{10}\left(2\times10^{-27}\right)$$

= 1.06

= 1 V

35. Assume organic compound is A and n molecules of A polymerise

$\Delta T_f = iK_f m$

$$i = \frac{\Delta T_f}{K_f m} = \frac{0.93\times94\times100}{1.86\times9.4\times1000}$$

$$i = \frac{1}{n} = 0.5$$

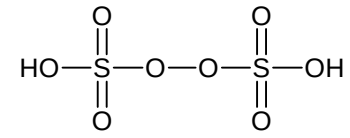
n = 2

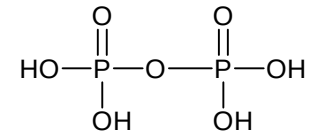
36. Average velocity $= \frac{5\times5+10\times3+6\times6}{21} = 4.33$

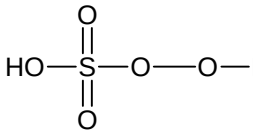
rms velocity $= \sqrt{\frac{5\times25+10\times9+6\times36}{21}} = 4.53$

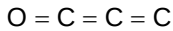
$\frac{V_{av}}{V_{rms}} = 0.955 = 9.5 \times 10^{-1}$

x = 1

37. 
Marshall acid a = 0


Pyrophosphoric acid b = 0


Caro acid c = 1


Carbon suboxide d = 2

38. $\frac{r_c}{r_a} = 0.858$

From above ratio it is clear that cation occupy cubic void in simple cubic unit cell, hence $\sqrt{3}a = 2(r_c + r_a)$

a = 4

40. $\underset{A}{FeCl_3} + 3KCN \longrightarrow \underset{B}{Fe(CN)_3} + 3KCl$

$Fe(CN)_3 + 3KCN \longrightarrow \underset{C}{K_3[Fe(CN)_6]}$

$4FeCl_3 + 3K_4[Fe(CN)_6] \longrightarrow \underset{D}{Fe_4[Fe(CN)_6]_3} + 12KCl$

$Fe_4[Fe(CN)_6]_3 + 12NaOH \longrightarrow 4Fe(OH)_3 + 3Na_4[Fe(CN)_6]$

$FeCl_3 + Na_2HPO_4 \longrightarrow \underset{E}{FePO_4} + 2NaCl + HCl$

# Mathematics

## PART – III

### SECTION – A

41. Put i = j = k
$a_{ij} = 0$ and put k = i $\Rightarrow a_{ij} = -a_{ji}$
So, matrix is skew symmetric of odd order

42. $y = (2x+3) + \dfrac{7}{(x+3)}$
x, y $\in$ I; x + 3 = $\pm 1, \pm 7$
(x, y) $\rightarrow$ (–2, 6), (–4, –12), (4, 12), (–10, –18)

43. $\left[a_{n+2} = 2a_{n+1} - 3a_n\right]$

44. $$\frac{x}{(x+i)!} \le \frac{x+i-1}{(x+i)!} = \frac{1}{(x+i-1)!} - \frac{1}{(x+i)!}$$
$$x\sum_{i=1}^{y} \frac{1}{(x+i)!} \le \frac{1}{x!} - \frac{1}{(x+y)!} \qquad \text{..... (1)}$$
$$\lim_{y\to\infty}\left(f(x,y)\right) \le \frac{1}{xx!} \le \frac{1}{x!} \qquad \text{..... (2)}$$
$$f(x,x) = \frac{1}{(x+1)!} + \frac{1}{(x+2)!} + ..... + \frac{1}{(x+x)!} \le \frac{x}{x!} = \frac{1}{(x-1)!}$$

45. A.M. $\ge$ H.M. for $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$, we get $\dfrac{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}{3} \ge \frac{3}{6}$
$$\Rightarrow \frac{1}{a}+\frac{1}{b}+\frac{1}{c} \ge \frac{3}{2}$$
Now, $$\frac{\left(a+\frac{1}{b}\right)^2 + \left(b+\frac{1}{c}\right)^2 + \left(c+\frac{1}{a}\right)^2}{3} \ge \left(\frac{a+\frac{1}{b}+b+\frac{1}{c}+c+\frac{1}{a}}{3}\right)^2 \ge \left(\frac{6+\frac{3}{2}}{3}\right)^2$$
$$\left(a+\frac{1}{b}\right)^2 + \left(b+\frac{1}{c}\right)^2 + \left(c+\frac{1}{a}\right)^2 \ge \frac{75}{4}$$

46. $P_n = P_{n-1} + P_{n-2}$, $P_1 = 1$, $P_2 = 1$
$P_3 = 1 + 1 = 2$
$P_4 = 2 + 1 = 3$
$P_5 = 5$
$P_6 = 8$
$P_7 = 13$
$P_8 = 21$
$P_9 = 34$
$P_{10} = 55$

47. (A) Intersection point is mid-point of AB and CD
(B) $2\alpha + 6\beta + 15\gamma = 7$

$\left(2\hat{i}+6\hat{j}+3\hat{k}\right)\cdot\left(\alpha\hat{i}+\beta\hat{j}+5\gamma\hat{k}\right)\le 7\times\sqrt{\alpha^2+\beta^2+25\gamma^2}$

$\left(2\alpha+6\beta+15\gamma\right)\le 7\sqrt{\alpha^2+\beta^2+25\gamma^2}$

$\alpha^2+\beta^2+25\gamma^2\ge 1$

(C) Triangle is equilateral

(D) Intersection points are (1, 2, 3) and (2, –3, 1)

48. Let $N = x_1x_2x_3 \ldots x_{10}$ be one of such number

$\sum x_i = 45$, hence N is divisible by 9. Hence N should by divisible by $11111 \times 9 = 99999$

Now $x_1x_2x_3 \ldots x_{10} = x_1x_2x_3x_4x_5 \times 10^5 + x_6x_7x_8x_9x_{10}$

$= x_1x_2x_3x_4x_5 \times 99999 + (x_1x_2x_3x_4x_5) + (x_6x_7x_8x_9x_{10})$

Now $x_1x_2x_3x_4x_5 < 99999$

$x_6x_7x_8x_9x_{10} < 99999$

$x_1x_2x_3x_4x_5 + x_6x_7x_8x_9x_{10} < 2 \times 99999$

$x_1x_2x_3x_4x_5 + x_6x_7x_8x_9x_{10} = 99999$

$x_1 + x_6 = x_2 + x_7 = x_3 + x_8 = x_4 + x_9 = x_5 + x_{10} = 9$

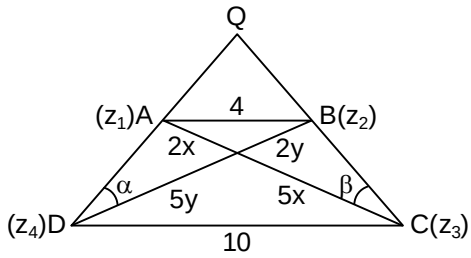
So, smallest number 1023489765

Largest number 9876501234

Total number $9 \times 8 \times 6 \times 4 \times 2 \times 1 \times 1 \times 1 \times 1 = 3456$

49.-50. $\Delta ABP$ and $\Delta CDP$ are similar if AP = 2x,

BP = 2y then CP = 5x, DP = 5y




Area of trapezium $ABCD = \frac{49}{2}xy$

$\tan\alpha = \frac{2x}{5y}, \tan\beta = \frac{2y}{5x}$ also $\alpha + \beta = 45^o$

$\Rightarrow xy = \frac{10}{21}\left(x^2+y^2\right)$

Also $AB^2 = AP^2 + BP^2 \Rightarrow x^2 + y^2 = 4 \Rightarrow xy = \frac{40}{21}$

$Ar(\Delta PCD) = 5xy$

51. $9 + 8 + 6 + 5 + 4 = 32$

Set P(A , B, C, D, E, F, G) = (2, 3, 4, 5, 6, 8, 9)

$n(S) = {}^7C_5 \times 5!$

Total numbers divisible by $5 = {}^6C_4 \times 4!$

Total numbers divisible by $3 = {}^3C_1 \times 5! + {}^3C_2 \times 5! = 6!$

Total numbers divisible by 4 = 720

52. 9, 8, 6, 5, 4

(A and B) ..... $9 + 8 + 6 + 5 + 4 = 32$

$|(9 + 8) – (6 + 5 + 4)| = 2$ not divisible by 11

$|(9 + 6) – (8 + 5 + 4)| = 3$ not divisible by 11

$|(9 + 5) – (8 + 6 + 4)| = 4$ not divisible by 11

$|(9 + 4) – (8 + 6 + 5)| = 6$ not divisible by 11

$|(8 + 6) – (9 + 5 + 4)| = 4$ not divisible by 11

$|(8 + 5) – (9 + 6 + 4)| = 6$ not divisible by 11

$|(8 + 4) – (9 + 6 + 5)| = 8$ not divisible by 11

$|(6 + 5) – (9 + 8 + 4)| = 10$ not divisible by 11

$|(6 + 4) – (9 + 8 + 5)| = 12$ not divisible by 11

$|(5 + 4) – (9 + 8 + 6)| = 14$ not divisible by 11

(C) ..... 4 8

$\underline{5}\ \underline{6}$
$\underline{6}\ \underline{4}$
$\underline{8}\ \underline{4}$
$\underline{9}\ \underline{6}$

Probability = $\frac{5\times 3!}{5!}=\frac{1}{4}$

## SECTION – C

53. Let $a + 2b + c = x$, $a + b + 2c = y$, $a + b + 3c = z$
$a = -x + 5y - 3z$, $b = z + x - 2y$, $c = z - y$
Apply A.M. $\geq$ G.M.
$$\frac{a+3c}{a+2b+c}+\frac{4b}{a+b+2c}-\frac{8c}{a+b+3c}\geq -17+12\sqrt{2}$$

54. $\alpha+\beta=\frac{a_2}{a_1}$, $\alpha\beta=\frac{a_3}{a_1}$, $\alpha, \beta \in (1, 2)$
$\Rightarrow \alpha - 1, \beta - 1, 2 - \alpha, 2 - \beta \in (0, 1)$
A.M. $\geq$ G.M. $\Rightarrow \frac{(\alpha-1)+(2-\alpha)}{2}\geq\sqrt{(\alpha-1)(2-\alpha)}$
$\Rightarrow (\alpha-1)(2-\alpha)\leq\frac{1}{4}$ similarly $(\beta - 1)(2 - \beta) \leq \frac{1}{4}$
$0<(\alpha-1)(2-\alpha)(\beta-1)(2-\beta)<\frac{1}{16}$ (Both can't equal to $\frac{1}{4}$ simultaneously ($\alpha \neq \beta$)
$$0<\left(\frac{a_3}{a_1}-\frac{a_2}{a_1}+1\right)\left(4-\frac{2a_2}{a_1}+\frac{a_3}{a_1}\right)<\frac{1}{16}$$
$$0<\frac{(a_1-a_2+a_3)(4a_1-2a_2+a_3)}{a_1^2}<\frac{1}{16}$$
$$\frac{a_1^2}{16}>(a_1-a_2+a_3)(4a_1-2a_2+a_3)$$
$\frac{a_1^2}{16} > f(1)\cdot f(2) \Rightarrow \frac{a^2}{16} > 1$
$\Rightarrow$ Least integral value of a is 5

55. Let $\frac{n}{m}$ be the common ratio of G.P., where n and m are coprime integers n > m
$a_5 = a_1 \cdot \frac{n^4}{m^4}$ so let $a_1 = km^4$ ($k \in I^+$)
G.P. $km^4$, $km^3n$, $km^2n^2$, $kmn^3$, $kn^4$
$kn^4 < 100 \Rightarrow$ n can't be greater than or equal to 4
**Case-I:** n = 3 and m = 2
$\Rightarrow a_5 = kn^4 < 100$
$\Rightarrow k < \frac{100}{81} \Rightarrow k = 1$
G.P. (16, 24, 36, 54, 81)
**Case-II:** n = 3 and m = 1
$\Rightarrow a_5 = 81k < 100 \Rightarrow k = 1$
G.P. (1, 3, 9, 27, 81)

**Case-III:** n = 2, m = 1

$\Rightarrow k < \frac{100}{16}$

$\Rightarrow$ k = 1, 2, 3, ..... 6

G.P.'s (1, 2, 4, 8, 16), (2, 4, 8, 16, 32), (3, 6, 12, 24, 48), (4, 8, 16, 32, 64), (6, 12, 24, 48, 96), (5, 10, 20, 40, 80)

56. Highest power of $x = \frac{n(n+1)}{2} = m$ (let) have to find coefficient of $x^{m-9}$

$-9 + (-8 \times -1) + (-7 \times -2) + (-6 \times -3) + (-5 \times -4) + (-6 \times -2 \times -1)$
$+ (-5 \times -3 \times -1) + (-4 \times -3 \times -2) = 0$

57. $\frac{\frac{a^4}{2}+\frac{a^4}{2}+\frac{b^6}{4}+\frac{b^6}{4}+\frac{b^6}{4}+\frac{b^6}{4}+\frac{c^8}{2}+\frac{c^8}{2}}{8} \geq \left(\frac{a^8b^{24}c^{16}}{2^{12}}\right)^{\frac{1}{8}}$

Equality hold for a = 1, $b = 2^{1/6}$, c = 1

58. $\Sigma|z + \alpha^k z^2|^2 = \Sigma|z|^2 + \Sigma|\alpha^k|\,|z^2|^2 + 0$

59. $AA^T = 4I \Rightarrow |A| = \pm 2$

$A^T = 4A^{-1} = \frac{4\text{Adj}(A)}{|A|}$

$$\Rightarrow \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} = \frac{4}{|A|}\begin{bmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{bmatrix}$$

Now, $a_{ij} = \frac{4}{|A|}\, c_{ij} = -2c_{ij} \Rightarrow |A| = -2$

Now, $|A + 4I| = |A + AA^T| = |A|\,|I + A^T| = -2|(I + A)^T| = -2|I + A|$ so, $\frac{|A+4I|}{|A+I|} = -2 = -5\lambda$

$\lambda = \frac{2}{5}$

60. $x^3 - x^2 - 1 = 0 \rightarrow \alpha, \beta, \gamma \Rightarrow \alpha^2 + \beta^2 + \gamma^2 = 1$

$\alpha^2 + \beta^3 + \gamma^4 = \alpha^2 + \beta^2 + 1 + \gamma^3 + \gamma = 3 + \gamma$

In given equation $x \rightarrow x - 3 \Rightarrow x^3 - 10x^2 + 33x - 37 = 0$